

Benchmarking Comparative Single-Cell Analysis of CMV Status

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1 Introduction

A clinical label describes a donor, whereas the informative signal may occur in only a small subset of that donor’s cells. CellCNN addresses this mismatch through supervised representation learning: shared cell filters and pooling are trained jointly to predict a sample-level outcome, allowing rare, informative responses to influence the representation [1]. This differs from first clustering cells and classifying their population frequencies, as in Citrus [3], or labelling every training cell with its donor’s outcome, as in a single-cell SVM. The paper also compares logistic regression, random forests, distance-based outlier detection, single-marker gates, moment summaries and a denoising autoencoder; Supplement S1 distinguishes these literature baselines from our implemented methods.

We study the supplied CMV cohort derived from Horowitz et al. [2]. CMV exposure is associated with expansions of NKG2C/CD57-expressing NK populations, but neither donor status nor a marker average establishes the identity of individual cells. We therefore ask three connected questions: how do dimensionality reduction and clustering describe the cell landscape, how accurately do the three classifiers predict unseen donors, and which model-associated cell subsets recur? A short final experiment examines modified CellCNN responses and pooling.

2 Methods

2.1 Dataset and preprocessing

The mass-cytometry (CyTOF) data comprise 20 donors (9 CMV-positive, 11 negative) and 37 protein markers. Technical channels are excluded. We use $a = \text{asinh}(x/5)$ and donor-balanced samples, with distinct analysis scopes: 40,000 `gated_NK` cells for dimensionality reduction (2,000/donor), 20,000 `gated_alive` PBMCs for clustering (1,000/donor), and `gated_alive` for classification. The latter contains 3,438,750 events. The interpretation map contains 10,000 cells (500/donor), but selected-cell profiles use full test donors. Exploratory marker scaling uses the corresponding analysis sample; classifier scaling is fitted only on training donors. Data checks and sampling details are in Supplement S1.

2.2 Dimensionality reduction

PCA retains 29 components explaining 91.50% of variance; these form the reference for t-SNE, UMAP and embedding-quality measurements. We sweep six t-SNE perplexities [4] and 64 UMAP settings [5] on 5,000 cells. Trustworthiness $T(15)$ [6] selects the configuration; for UMAP, learning rate is first selected by mean T across two fixed configurations, followed by neighbour count and minimum distance. Selected settings are fitted to all 40,000 cells. For direct comparison of every pair of final maps we use Procrustes disparity M^2 , the residual after optimal translation, rotation/reflection and scaling [9]. Lower values mean closer aligned shapes. Additional neighbourhood and distance measures appear in Supplement S2.

2.3 Clustering and biological interpretation

We cluster standardised 37-marker PBMC profiles, not their two-dimensional projections, using established implementations [9, 10]. The 44 configurations cover k-means, four hierarchical linkages and Leiden communities [8]. Ten repeats retain 80% of cells within each donor, refit scaling and clustering, and match clusters by their largest Jaccard overlap on retained cells. Stability is the mean of cluster-wise median overlaps. Within each method/linkage, we minimise Davies–Bouldin (DB) [7] among configurations with stability ≥ 0.6 . This heuristic criterion is complemented by marker profiles: (cluster median – sample median)/sample IQR after ArcSinh transformation. Settings and cluster-wise stability are reported in Supplement S2.

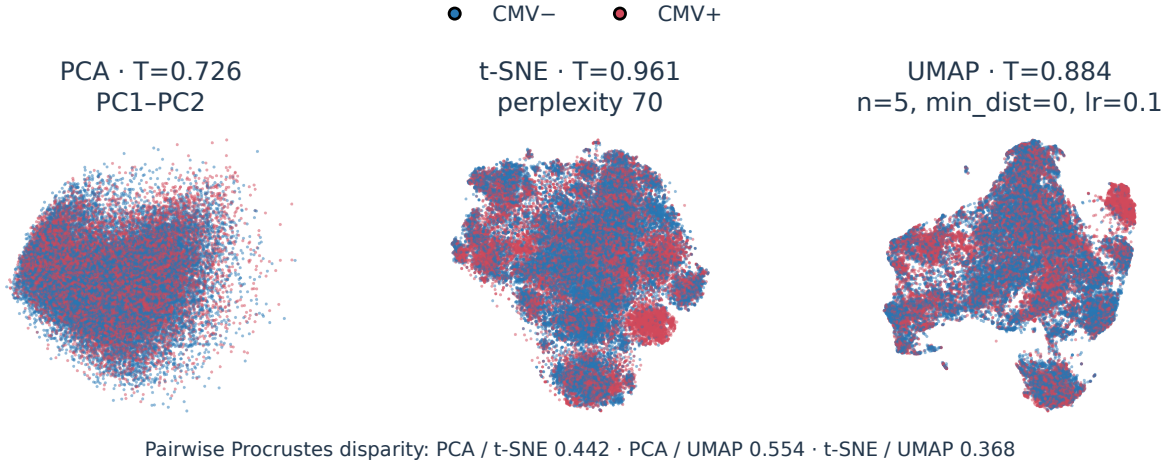


Figure 1: The same 40,000 NK-gated cells in the three selected embeddings, coloured by donor CMV status. $T(15)$ measures neighbourhood preservation relative to 29 PCs. The three direct pairwise comparisons use Procrustes disparity; orientation and scale are aligned before comparison.

2.4 Comparative single-cell classification

The benchmark uses 30 shared donor splits: 14 training donors (7 per class), six test donors (2 positive, 4 negative), and three inner folds for model selection. Test donors never enter training fits. **CellCNN** uses 3–5 linear/ReLU filters, means of the top 1% responses and a softmax output. Our PyTorch implementation retains the selected inner-fold network, trained on 9 or 10 donors, and averages five test-input probabilities. **Citrus** uses its original R implementation: Ward clusters, cluster-frequency features and L1-logistic regression. **SVM** uses a linear cell scorer trained with donor labels; its donor score averages the highest 1% of full-donor margins. Citrus and SVM are refitted on all 14 training donors. Our SVM aggregation and fixed CellCNN filter search are project adaptations. Full implementation and selection details are in Supplement S3.

2.5 Identification of model-associated subsets

We focus on the CMV-positive model direction. For active CellCNN filters with positive output contrast, $q_i = \max_f r_{if}/M_f$ selects $q_i > 0.5$, where M_f is the original training-reference maximum. SVM selects positive centred margins $m_i - \tau > 0$ within the donor’s top 1%, using its inner-trained threshold. For Citrus, $q_i = \max(\{0\} \cup \{\beta_j : i \in A_j\})$ selects $q_i > 10^{-10}$; membership A_j is inherited from the nearest original training cell. Each cell counts once per split. Its displayed frequency divides selections by all test appearances of its donor, including empty selections. Marker profiles average selected cells within nonempty test visits, visits within donors, then donors equally. Equations and the verified Citrus reconstruction are given in Supplement S3.

3 Results

3.1 Dimensionality reduction

The first two PCs explain only 20.29% of variance. t-SNE perplexity 70 gives the highest sweep trustworthiness (0.9271); values at perplexities 30–100 differ modestly. Low perplexity emphasises very local arrangements, whereas higher values change the scale of visible neighbourhoods. For UMAP, smaller minimum distance packs points more tightly and neighbour count changes local connectivity; learning rate also affects the result (Supplement S2). The selected setting is 5 neighbours, minimum distance 0 and learning rate 0.1.

On the final 40,000-cell maps, t-SNE has the highest T (0.9608), ahead of UMAP (0.8838) and PCA (0.7258). PCA preserves pairwise distances better in the 2,000-cell distance sample ($r = 0.6204$, versus 0.4799 and 0.4317). Thus local and global criteria favour different methods. t-SNE and UMAP are the most similar pair under Procrustes ($M^2 = 0.3683$; Fig. 1). Donor CMV classes remain substantially intermixed; a visible island alone is neither a cell-type annotation nor a classification result.

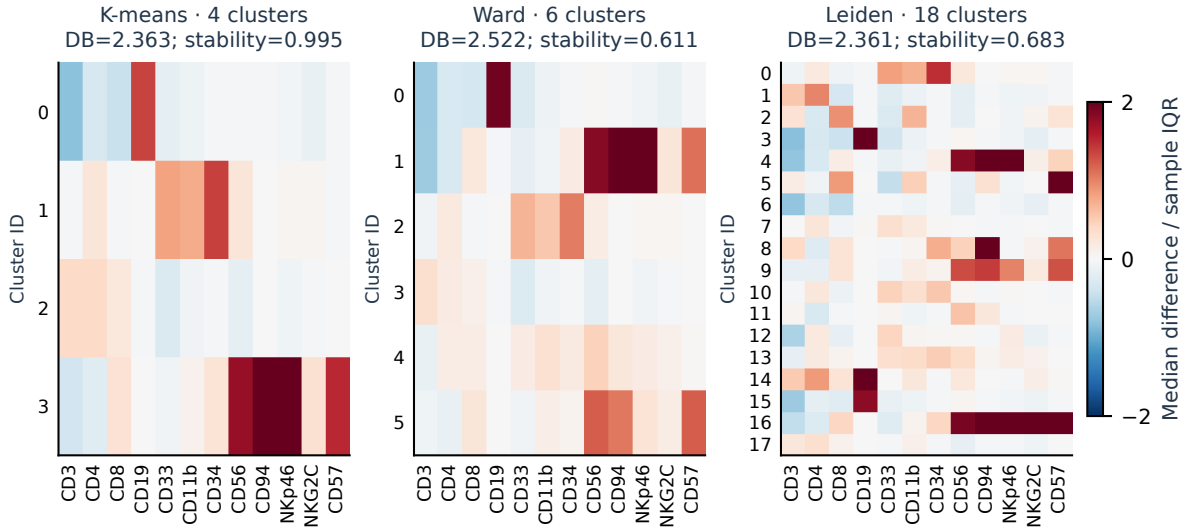


Figure 2: Marker evidence for the selected k-means, Ward and Leiden solutions on 20,000 live PBMCs. Twelve markers share one relative-IQR scale; colours are clipped at ± 2 . Cluster IDs are local to each method. DB is evaluated in 37D; stability comes from the original ten cell-subsampling repeats. Full profiles appear in Supplement S2.

3.2 Clustering and biological interpretation

K-means selects four broad groups with mean stability 0.995. Ward yields six groups (0.611), and Leiden 18 (0.683); the DB scores of k-means and Leiden are nearly equal (Fig. 2). Marker evidence is informative at several resolutions: the CD19-enriched k-means group is B-compatible, while its Nkp46/CD94-enriched group is NK-associated. Leiden separates a strongly NKG2C-enriched small group, illustrating the additional detail available at finer resolution. These descriptions concern relative marker medians, not validated cell-type labels or proof of single-cell co-expression.

The best single-linkage score is misleading biologically: it separates 19,999 cells from one event despite $DB = 0.264$ and $stability = 1$. Moreover, three of six Ward clusters and four of 18 Leiden clusters have stability below 0.6 even though their method averages pass. K-means is therefore the strongest reproducible coarse description here; Ward and Leiden offer subdivisions requiring further validation. No universally superior biological partition is established without reference annotations.

3.3 Classification benchmark

CellCNN and SVM both achieve median test AUC 0.875, compared with 0.625 for Citrus (Fig. 3). CellCNN exceeds SVM in 12 splits, ties in 12 and falls below it in six; its mean paired AUC advantage is only 0.0167. Against Citrus, the corresponding counts are 22/6/2. Median balanced accuracy is 0.625 for CellCNN/SVM and 0.500 for Citrus, showing that good rankings do not automatically produce strong thresholded decisions. Complete AUC, average-precision and balanced-accuracy distributions are in Supplement S5.

3.4 Model-associated cell subsets

The common test-cell analysis reveals different selection patterns (Fig. 4). CellCNN, SVM and Citrus select 863, 93 and 1,583 map cells at least once, but only 84, 9 and 11 in at least half of their donor’s test appearances. CellCNN has a consistent core within a broader, less stable selection; Citrus marks many cells intermittently. CellCNN/SVM profiles show stronger NKG2C enrichment (mean reference z-scores 2.862/3.004) than Citrus (0.972), alongside NK-associated markers. This is compatible with the CMV-related phenotype described by Horowitz et al. [2], but the pooled profiles can mix populations. Different selection rules prevent interpreting cell counts alone as biological superiority.

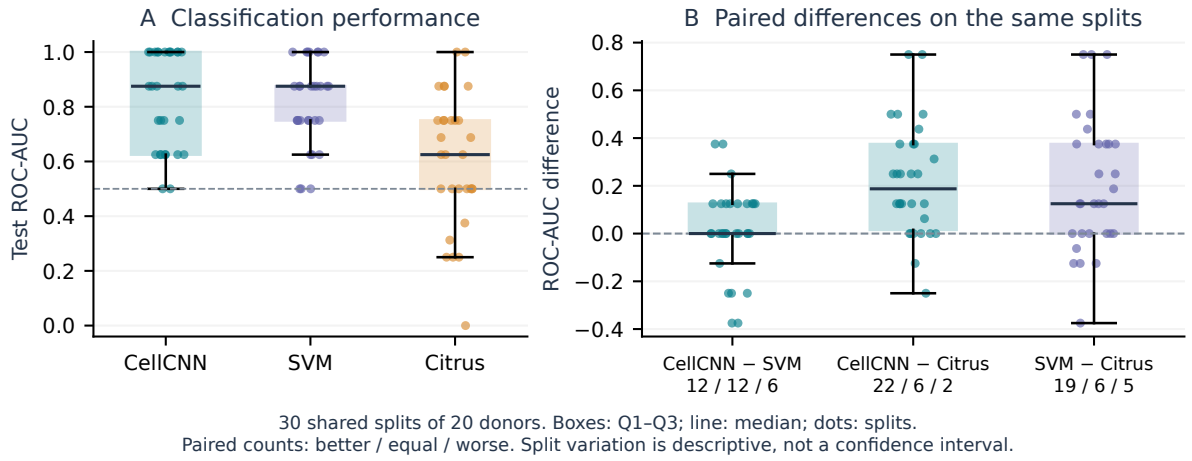


Figure 3: Performance on 30 shared donor splits. A: test AUC; B: paired AUC differences on identical splits. Boxes show Q1-Q3, lines medians and dots splits. Counts denote better/equal/worse performance of the first method. These overlapping evaluations are descriptive, not independent cohorts or confidence intervals.

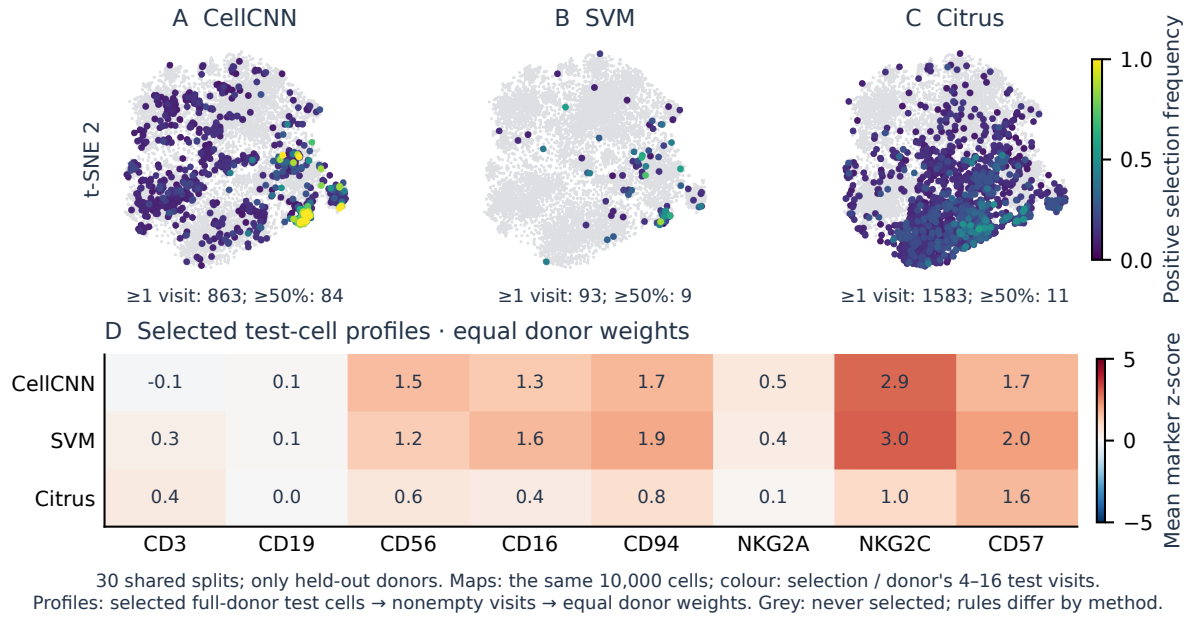


Figure 4: A-C: selections on the same 10,000-cell `gated_alive` t-SNE reference (perplexity 30), evaluated only during each donor's 4-16 test appearances. Grey means never selected. D: selected full-test-donor marker profiles with equal weights for all 20 contributing donors, on a common z-scale. Empty selections remain in frequency denominators but have no marker profile.

3.5 Exploratory extension: modified CellCNN

We add diagonal quadratic terms, $\text{ReLU}(b_f + w_f^T z + q_f^T z^{\odot 2})$, to capture nonlinear marker responses without cross-marker products. The quadratic weights start at zero, so the hard-pooling variant contains the linear baseline at initialisation. We compare hard top-1% pooling with a learnable fraction that smoothly weights sorted responses, and include a diagonal Mahalanobis/ReLU prototype variant (Supplement S4). The latter constrains responses to neighbourhoods of learned centres and uses a different regularisation scheme; its result is therefore not an isolated test of distance geometry. On 100 shared splits, both quadratic variants increase mean AUC by 0.1050; soft pooling adds no further mean AUC gain. The prototype variant performs worse. These are exploratory comparisons on the same small donor cohort.

Model	Mean network AUC (100 splits)
CellCNN baseline	0.8075
Quadratic Top-1%	0.9125
Quadratic Soft- α	0.9125
Mahalanobis/ReLU Soft- α	0.7375

4 Discussion and conclusion

The analyses answer complementary questions: embedding quality concerns geometry, clustering stability concerns reproducibility within the sampled cells, and held-out classification concerns donor prediction. None alone validates a biological cell type. Twenty donors, including related individuals, repeated overlapping splits and heuristic subset thresholds limit generalisation. Joint exploratory projections are also distinct from independent biological validation. CellCNN and SVM give similar rankings, while their cell selections differ; Citrus is weaker in this benchmark. Quadratic filters are a promising extension, but confirmation requires an independent donor cohort and biological assessment of the selected cells.

References

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